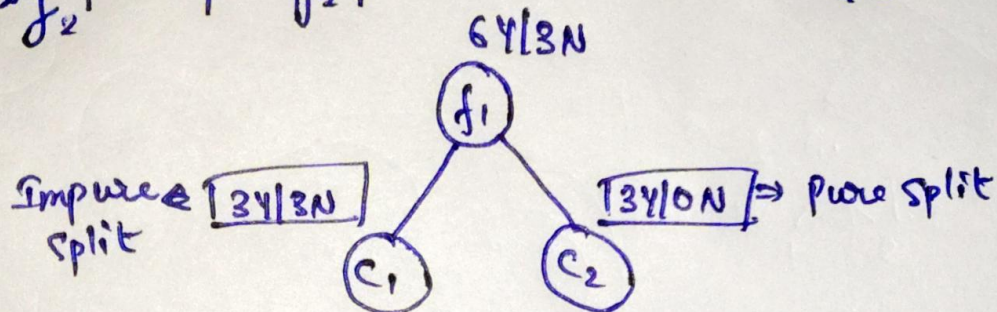


1. Entropy

$$H(S) = -p_+ \log_2 p_+ - p_- \log_2 p_-$$



$$H(c_1) = -p_+ \log_2 p_+ - p_- \log_2 p_-$$

$$= -\frac{8}{6} \log_2 \frac{8}{6} - \frac{2}{6} \log_2 \frac{2}{6}$$

$$= 1 \Rightarrow \text{Impure Split}$$

$$H(c_2) = -\frac{3}{3} \log_2 \frac{3}{3} - 0 \log_2 0$$

$$= -1 \log_2 1 \Rightarrow 0 \Rightarrow \text{Pure Split}$$

2. Gini Impurity

$$G.I = 1 - \sum_{i=1}^n (p_i)^2$$

$$G.I = 1 - \sum_{i=1}^n (p_i)^2$$

$$= 1 - ((p_+)^2 + (p_-)^2)$$

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$$= 1 - \left(\left(\frac{1}{2} \right)^2 + \left(\frac{1}{2} \right)^2 \right)$$

$$= 0.5 \Rightarrow \text{Impure Split}$$

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$$= 1 - \left(\left(\frac{3}{3} \right)^2 \right)$$

$$= 1 - 1 = 0 \Rightarrow \text{Pure Split}$$

